Γ-only real-space RI *GW* for two-dimensional materials: working equations from the atomic orbitals on the grid to the correlation self-energy

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Abstract

We give, in the order in which they are evaluated, every equation of the Γ-only large-cell *GW* scheme with real-space resolution of the identity (RI-RS) for a two-dimensional insulating layer. The screened interaction is computed at a single reciprocal-space point, $\mathbf{q} = 0$, where the real-space lattice sum of the bare Coulomb operator between charged auxiliary functions diverges. The divergence is removed exactly as a rank-one constant, the charge channel is carried analytically by one additional basis function, the normalized constant $\phi_0 = 1/\sqrt{A}$, and its contribution to the correlation part of the screened interaction is integrated analytically over the whole small Brillouin zone of the supercell (head-and-wing correction, following J. Wilhelm, *Periodic real-space RI for full-k GW band structures*, Sec. VI). Nothing is subtracted, because the single sampled point carries no head after the constant has been removed. Validation on a 14×14 hexagonal boron nitride cell (392 atoms) gives a G_0W_0 gap of 6.699 eV against 6.595 eV (shifted 2×2 mesh) and 6.582 eV (4×4) at 64 % of the cost of the 2×2 calculation.

1 Setting and notation

The simulation cell is a layer with in-plane lattice vectors $\mathbf{a}_1, \mathbf{a}_2$, periodic in x and y only, with a non-periodic direction z. Its layer area is

$$A = |\mathbf{a}_1 \times \mathbf{a}_2|, \tag{1}$$

and the in-plane reciprocal vectors $\mathbf{b}_1, \mathbf{b}_2$ satisfy $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$. The vacuum height never enters any equation below: all Fourier transforms average over a layer cell, not over a three-dimensional cell that mostly contains vacuum.

Indices: $\mu, \nu, \lambda, \sigma$ label atomic orbitals (AOs) φ_μ; P, Q, S label auxiliary (RI) Gaussians ϕ_P; ℓ, ℓ' label points $\mathbf{r}_\ell$ of a real-space grid; i labels occupied and a empty Kohn–Sham states; n labels any state; $\mathbf{R}$ labels lattice vectors; τ is imaginary time and ω imaginary frequency. $\alpha, \beta \in \{x, y, z\}$ are Cartesian directions and $\hat{\mathbf{q}}$ an in-plane unit vector. All quantities are real: there is one real k-point, Γ, and the single complex factor of the construction (the wing of W) is split off analytically.

The RI metric is a short-ranged kernel v_ω, the truncated Coulomb operator $v_{\mathrm{tc}}(r) = \theta(r_c - r)/r$ in the calculations reported here. Its zero-wavevector value is

$$\int \mathrm{d}\mathbf{r}\, v_\omega(r) = \begin{cases} \pi/\omega^2 & \text{erfc-attenuated metric,} \\ 2\pi r_c^2 & \text{truncated Coulomb metric,} \end{cases} \tag{2}$$

and is referred to as the metric head.

Imaginary time and frequency are discretized on minimax grids $\{\tau_j\}$, $\{\omega_k\}$, $j, k = 1, \ldots, N$. The cosine and sine transforms are written with their minimax weights,

$$f(\mathrm{i}\omega_k) = \sum_j \gamma^{\cos}_{kj} f(\mathrm{i}\tau_j), \qquad f(\mathrm{i}\tau_j) = \sum_k \tilde{\gamma}^{\cos}_{jk} f(\mathrm{i}\omega_k), \qquad f(\mathrm{i}\omega_k) = \sum_j \gamma^{\sin}_{kj} f(\mathrm{i}\tau_j), \tag{3}$$

where the first represents $f(\mathrm{i}\omega) = 2\int_0^\infty \mathrm{d}\tau \cos(\omega\tau) f(\mathrm{i}\tau)$ for an even function of τ, the second its inverse, and the third $f(\mathrm{i}\omega) = 2\mathrm{i}\int_0^\infty \mathrm{d}\tau \sin(\omega\tau) f(\mathrm{i}\tau)$ for an odd one.

2 Atomic orbitals on the real-space grid

Modified Lebedev atomic grids are placed at every atom and their union is the global grid $\{\mathbf{r}_\ell\}_{\ell=1}^{N_{\mathrm{grid}}}$, all points in the home cell. Because there is only one k-point, the AO values on the grid are the Γ Bloch sums over periodic images,

$$\Phi_\mu(\mathbf{r}_\ell) = \sum_{\mathbf{R}} \varphi_\mu(\mathbf{r}_\ell - \mathbf{A}_\mu - \mathbf{R}), \tag{4}$$

where $\mathbf{A}_\mu$ is the center of φ_μ. The matrix $\Phi_{\ell\mu}$ is sparse: $\Phi_{\ell\mu}$ is kept only if $|\mathbf{r}_\ell - \mathbf{A}_\mu - \mathbf{R}|$ is within the AO radius for some image. At the 14×14 h-BN cell 24.9 % of its elements are nonzero.

3 Real-space RI: the fit coefficients $Z_{\ell P}$

RI-RS expresses products of AOs through the grid. Define the grid metric and the grid–RI right-hand side,

$$D_{\ell\ell'} = \Big[\sum_{\mu} \Phi_\mu(\mathbf{r}_\ell)\Phi_\mu(\mathbf{r}_{\ell'})\Big]^2, \tag{5}$$

$$d_{\ell P} = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell)\,(\mu\nu|P)_\omega, \qquad (\mu\nu|P)_\omega = \sum_{\mathbf{R}\mathbf{R}'} \iint \mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}'\, \varphi_\mu^{\mathbf{R}}(\mathbf{r})\varphi_\nu^{\mathbf{R}'}(\mathbf{r})\, v_\omega(|\mathbf{r}-\mathbf{r}'|)\,\phi_P(\mathbf{r}'). \tag{6}$$

Writing $B_{(\mu\nu),\ell} = \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell)$ one has $D = B^{\mathrm{T}}B$ and $d_{\cdot P} = B^{\mathrm{T}} g_P$ with $g_{(\mu\nu),P} = (\mu\nu|P)_\omega$, so the normal equations

$$\sum_{\ell'} D_{\ell\ell'} Z_{\ell' P} = d_{\ell P} \tag{7}$$

are the least-squares solution of $\sum_\ell \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell) Z_{\ell P} \approx (\mu\nu|P)_\omega$: the coefficient $Z_{\ell P}$ is a quadrature weight times the value at $\mathbf{r}_\ell$ of the metric potential of ϕ_P. The solve is balanced and regularized,

$$j_\ell = \frac{1}{\max(D_{\ell\ell}, \sqrt{\epsilon_{\mathrm{floor}}})}, \qquad \big[j_\ell D_{\ell\ell'} j_{\ell'} + \lambda\,\delta_{\ell\ell'}\big] Z'_{\ell' P} = j_\ell d_{\ell P}, \qquad Z_{\ell P} = j_\ell Z'_{\ell P}, \tag{8}$$

with the Tikhonov parameter $\lambda = 10^{-8}$ and $\epsilon_{\mathrm{floor}} = 10^{-16}$. Because $d_{\ell P}$ decays away from the center of ϕ_P, Eq. (7) is solved independently for each atom P on the grid points inside a sphere around it (radius $r_{\mathrm{RI}} + r_{\mathrm{AO}}$), by a dense Cholesky factorization of the sphere's block of D. This sphere locality is what makes RI-RS linear scaling; it is valid precisely because every column $d_{\ell P}$ is localized, a point that becomes important in Sec. 8.1.

Every subsequent contraction between the grid and the RI basis uses the same matrix:

$$X_{PQ} = \sum_{\ell\ell'} Z_{\ell P}\, X_{\ell\ell'}\, Z_{\ell' Q}, \qquad X_{\ell\ell'} = \sum_{PQ} Z_{\ell P}\, X_{PQ}\, Z_{\ell' Q}. \tag{9}$$

4 Green's functions and polarizability at Γ

With Kohn–Sham coefficients $C_{\mu n}$, eigenvalues ε_n and the Fermi level ε_F placed mid-gap, the occupied and empty Green's functions in imaginary time are

$$G^{\mathrm{occ}}_{\mu\nu}(\mathrm{i}|\tau|) = \sum_{i}^{\mathrm{occ}} C_{\mu i}\, \mathrm{e}^{-|(\varepsilon_i - \varepsilon_F)\tau|}\, C_{\nu i}, \qquad G^{\mathrm{vir}}_{\mu\nu}(\mathrm{i}|\tau|) = \sum_{a}^{\mathrm{vir}} C_{\mu a}\, \mathrm{e}^{-|(\varepsilon_a - \varepsilon_F)\tau|}\, C_{\nu a}, \tag{10}$$

$$G^{\mathrm{occ/vir}}_{\ell\ell'}(\mathrm{i}|\tau|) = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell)\, G^{\mathrm{occ/vir}}_{\mu\nu}(\mathrm{i}|\tau|)\, \Phi_\nu(\mathbf{r}_{\ell'}), \tag{11}$$

each built as $(\tilde{C})(\tilde{C})^{\mathrm{T}}$ with $\tilde{C}_{\mu n} = C_{\mu n} \mathrm{e}^{-|(\varepsilon_n - \varepsilon_F)\tau|/2}$. The independent-particle polarizability is an element-wise (Hadamard) product on the grid, contracted to the RI basis with Eq. (9),

$$\chi_{\ell\ell'}(\mathrm{i}\tau) = G^{\mathrm{occ}}_{\ell\ell'}(\mathrm{i}|\tau|)\, G^{\mathrm{vir}}_{\ell\ell'}(\mathrm{i}|\tau|), \qquad \chi_{PQ}(\mathrm{i}\tau) = \sum_{\ell\ell'} Z_{\ell P}\, \chi_{\ell\ell'}(\mathrm{i}\tau)\, Z_{\ell' Q}. \tag{12}$$

The grid product is evaluated in panels of grid rows; a geometric cutoff on $|\mathbf{r}_\ell - \mathbf{r}_{\ell'}|$ limits the pairs. At Γ, $\chi_{\ell\ell'}$ is a Gram matrix of occupied–empty products and therefore exactly positive semidefinite; no clipping of negative eigenvalues is needed. It is transformed to imaginary frequency with the cosine weights of Eq. (3), $\chi_{PQ}(\mathrm{i}\omega_k) = \sum_j \gamma^{\cos}_{kj} \chi_{PQ}(\mathrm{i}\tau_j)$.

5 The screened interaction at Γ: the problem and the plan

W is needed at one point only, $\mathbf{q} = 0$. There the bare Coulomb matrix between charged auxiliary functions diverges. The scheme handles this in three steps:

1. The divergence sits in one direction of the auxiliary basis, the charge vector c. It is removed from $V(0)$ as a single constant, $V(0) \to V(0) - D_{\mathrm{box}}\, cc^{\mathrm{T}}$ (Sec. 6).
2. ε and W are computed from this finite $V(0)$ in the usual way (Sec. 7). What is now missing from W is exactly the screened interaction in the charge channel.
3. That missing piece is added back analytically: the charge channel is represented by the constant function $\phi_0 = 1/\sqrt{A}$, its contribution to W (the "head" and "wings") is known in closed form as a function of $\mathbf{q}$, and it is integrated over the whole small Brillouin zone of the supercell (Sec. 8). Nothing has to be subtracted, because step 1 already removed it from the sampled point.

6 Step 1: the bare Coulomb matrix at $\mathbf{q} = 0$

6.1 What is computed

The Coulomb matrix between two auxiliary functions ϕ_P and ϕ_Q in a periodic layer is the interaction of ϕ_P in the home cell with ϕ_Q in *every* cell. The code sums over a finite box of cells,

$$V^{\mathrm{lat}}_{PQ}(0) = \sum_{\mathbf{R}\in\mathcal{B}} \underbrace{\iint \mathrm{d}\mathbf{r}\, \mathrm{d}\mathbf{r}'\, \frac{\phi_P(\mathbf{r})\, \phi_Q(\mathbf{r}' - \mathbf{R})}{|\mathbf{r} - \mathbf{r}'|}}_{(P0|Q\mathbf{R})}, \qquad \mathcal{B} = \{n_1\mathbf{a}_1 + n_2\mathbf{a}_2 : |n_1|, |n_2| \le m\}, \tag{13}$$

with $m = (3^{s-1} - 1)/2$ and s the input `SIZE_LATTICE_SUM` ($s = 4$: $m = 13$, i.e. 27×27 cells). There are no images in the direction normal to the layer.

6.2 Far images see only the charge

Take one term $(P0|Q\mathbf{R})$ with $|\mathbf{R}|$ much larger than the extent of the two Gaussians. Seen from far away, ϕ_Q looks like a point charge. Its charge is its integral,

$$c_Q = \int \mathrm{d}\mathbf{r}\, \phi_Q(\mathbf{r}), \tag{14}$$

and the same for ϕ_P. The multipole expansion of the term is therefore

$$(P0|Q\mathbf{R}) = \frac{c_P\, c_Q}{|\mathbf{R}|} + \frac{(\text{dipole terms})}{|\mathbf{R}|^2} + \frac{(\ldots)}{|\mathbf{R}|^3} + \cdots \tag{15}$$

Summed over the box, the dipole terms cancel between $\mathbf{R}$ and $-\mathbf{R}$ and the $|\mathbf{R}|^{-3}$ and higher terms converge. The only part of the sum that does not settle down is the first one:

$$V^{\mathrm{lat}}_{PQ}(0) = c_P\, c_Q \sum_{\mathbf{R}\in\mathcal{B}} \frac{1}{|\mathbf{R}|} \; + \; (\text{a part that converges as the box grows}). \tag{16}$$

6.3 Why the first part diverges

In two dimensions the number of lattice points in a ring of radius R and width $\mathrm{d}R$ is $2\pi R\,\mathrm{d}R/A$ (one point per area A). Each of them contributes $1/R$, so the ring contributes $2\pi\,\mathrm{d}R/A$ — the same for every ring. The sum therefore grows *linearly* with the box radius and never converges:

$$\sum_{\mathbf{R}\in\mathcal{B}} \frac{1}{|\mathbf{R}|} \;\approx\; \frac{1}{A}\int_{\mathcal{P}} \frac{\mathrm{d}^2 r}{|\mathbf{r}|} \;\propto\; \frac{2\pi}{A}\,R_{\mathrm{box}}. \tag{17}$$

On the h-BN cell this term is 7.4×10^4 Ha in the charge direction (defined next), against a few Ha for everything else.

6.4 The charge channel

Collect the charges into one vector $c=(c_1,\dots,c_{N_{\mathrm{RI}}})$ and normalize it, $\hat{c}=c/\|c\|$. Equation (16) says the divergent part of the matrix is $c\,c^{\mathrm{T}}$ times a number: it is a *rank-one* matrix, nonzero only along the single direction $\hat{c}$ of the auxiliary basis. We call that direction the *charge channel*. Any matrix can be split into its charge-channel part and the rest,

$$V = \underbrace{(\hat{c}^{\mathrm{T}}V\hat{c})\,\hat{c}\hat{c}^{\mathrm{T}}}_{\text{charge channel}} + \underbrace{\big(V-(\hat{c}^{\mathrm{T}}V\hat{c})\,\hat{c}\hat{c}^{\mathrm{T}}\big)}_{\text{rest}}, \tag{18}$$

and the whole problem of $\mathbf{q}=0$ is that the number $\hat{c}^{\mathrm{T}}V\hat{c}$ is infinite. Two remarks that matter later. First, not every auxiliary function is charged: an s-type Gaussian is, a p-type one is not, and a combination like $\phi_1-(c_1/c_2)\phi_2$ is neutral. Second, the charge channel also contains a *finite* part — the interaction of a charged function with its own neighbourhood, the second term of Eq. (16) projected onto $\hat{c}$ — which is physical and must be kept. Only the divergent constant may go.

6.5 D_{box}: the constant that is removed

The divergent number is the lattice sum of $1/|\mathbf{R}|$ over the box. Replacing the sum by the integral over the region $\mathcal{P}$ that the summed cells cover — the parallelogram $\{u_1\mathbf{a}_1+u_2\mathbf{a}_2 : |u_1|,|u_2|\le F\}$ with $F=3^{s-1}/2$, one cell per lattice point — defines

$$D_{\mathrm{box}} = \frac{1}{A}\int_{\mathcal{P}} \frac{\mathrm{d}^2 r}{|\mathbf{r}|}. \tag{19}$$

In polar coordinates $\mathrm{d}^2 r = r\,\mathrm{d}r\,\mathrm{d}\theta$, the $1/r$ cancels, and the radial integral is just the distance $r_{\max}(\theta)$ from the origin to the edge of the parallelogram in direction θ:

$$D_{\mathrm{box}} = \frac{1}{A}\int_0^{2\pi} \mathrm{d}\theta\; r_{\max}(\theta) = \frac{1}{A}\oint_{\partial\mathcal{P}} \frac{\mathbf{r}\times\mathrm{d}\mathbf{r}}{|\mathbf{r}|}, \tag{20}$$

a one-dimensional integral along the four edges of the box, done by Gauss–Legendre quadrature. It is a pure number fixed by the cell shape and the box size: 2.80 for the 6 Å water cell at $s=3$, 1.52 for the h-BN cell at $s=4$.

6.6 The Coulomb matrix used for W

Subtracting the constant along the charge direction, with the *unnormalized* charges so that the prefactor of Eq. (16) is matched,

$$V_{PQ}(0) = V^{\mathrm{lat}}_{PQ}(0) - D_{\mathrm{box}}\,c_P\,c_Q. \tag{21}$$

This removes exactly the first term of Eq. (16) and nothing else. What is left in the charge channel is small and finite: $\hat{c}^{\mathrm{T}}V\hat{c}$ goes from 796 to 8.9 Ha for the water cell and from 7.4×10^4 to -139 Ha

for h-BN. In reciprocal space this is the familiar "omit the $\mathbf{G} = 0$ term" of the Coulomb sum; Eq. (21) is its real-space form.

What one must *not* do is project the charge channel out altogether, $V \to \Pi V \Pi$ with $\Pi = 1 - \hat{c}\hat{c}^{\mathrm{T}}$. That would also discard the finite part of the charge channel (the second remark above), which step 3 does not put back, and it was measured to inflate every Σ^c matrix element by a factor of 5–9 on the water test.

7 Step 2: ε and W from the finite $V(0)$

With the RI metric $M_{PQ} = (P|v_\omega|Q)$ and V from Eq. (21),

$$\varepsilon(\mathrm{i}\omega) = 1 + V^{1/2} M^{-1} \chi(\mathrm{i}\omega) M^{-1} V^{1/2}, \tag{22}$$

$$W^c(\mathrm{i}\omega) = V^{1/2} \big(\varepsilon^{-1}(\mathrm{i}\omega) - 1\big) V^{1/2}, \tag{23}$$

$$W^{\mathrm{MIC}}(\mathrm{i}\omega) = M^{-1}\, W^c(\mathrm{i}\omega)\, M^{-1}. \tag{24}$$

This is the only k-point, so W^{MIC} is Eq. (24) with weight one. (The code's two-mesh extrapolation must be switched off for this; with it, Γ would enter W with weight $1/(1-\sqrt{2})$ in some matrix elements and weight 0 in others.) W^{MIC} is transformed to imaginary time and brought to the grid,

$$W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) = \sum_{PQ} Z_{\ell P}\, W^{\mathrm{MIC}}_{PQ}(\mathrm{i}\tau)\, Z_{\ell' Q}. \tag{25}$$

8 Step 3: the charge channel, added back analytically

8.1 The constant function and its auxiliary charges

The charge channel is represented by one extra basis function, the normalized constant,

$$\phi_0(\mathbf{r}) = \frac{1}{\sqrt{A}}, \qquad V_{00}(\mathbf{q}) = \frac{2\pi}{q}. \tag{26}$$

$2\pi/q$ is the exact two-dimensional Coulomb head; it is finite for every $q > 0$ and the $1/q$ singularity is integrable in two dimensions, which is why integrating over the Brillouin zone works where sampling at $q = 0$ does not.

At a small but finite $\mathbf{q}$ the charge channel is the Bloch-modulated constant $\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}}/\sqrt{A}$, and a fitted density $\rho = \sum_P \rho_P \phi_P$ couples to it through its Fourier component,

$$\langle \phi_{0,\mathbf{q}} | \rho \rangle = \frac{\hat{\rho}(\mathbf{q})}{\sqrt{A}} = \frac{1}{\sqrt{A}} \sum_P \rho_P\, \hat{\phi}_P(\mathbf{q}) \simeq \frac{1}{\sqrt{A}} \sum_P \rho_P\, c_P\, \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R}_P}, \qquad c_P = \int \mathrm{d}\mathbf{r}\, \phi_P(\mathbf{r}), \tag{27}$$

with $\mathbf{R}_P$ the centre of the auxiliary Gaussian ϕ_P. The last step is the point-charge limit $q_{\max}\sigma_P \ll 1$ of the Gaussian form factor; only s-type functions carry a charge. So the charge channel needs no fit of its own: the auxiliary charges c_P and the atomic positions are everything, and the correction is applied in the auxiliary basis (Sec. 8.5), reaching the grid through the same Z as every other part of W.

An earlier version gave ϕ_0 grid coefficients $Z_{\ell 0}$ from the fit of Eq. (7) with the constant as the target (solved globally by conjugate gradients, since its right-hand side does not decay) and applied the correction per grid pair, $Z_{\ell 0} Z_{\ell' 0} K_{\mathrm{h}}(\mathbf{R}_{\ell\ell'})$. That is numerically unusable. The RI-RS interpolation functions ξ_ℓ far from the atoms carry fit weights of 10^4–10^8 of alternating sign (on the 3×3 h-BN test cell, $\max|Z_{\ell 0}| = 2 \times 10^4$ at points with a density of 10^{-9}). They cancel exactly in linear functionals of a density, $\sum_\ell \rho(\mathbf{r}_\ell) Z_{\ell 0}$ gives $1/\sqrt{A}$ for a diagonal pair and 3×10^{-6} for a transition density, but not once each point is weighted by a different kernel value $K(\mathbf{R}_{\ell\ell'})$: the odd wing came out wrong by orders of magnitude and split symmetry-equivalent states by more than 1 eV, the even head survived only on large cells where the kernel is nearly flat over neighbouring points.

8.2 The neutral Gaussians

The Gaussians are made neutral by projecting out the charge direction,

$$\Pi = 1 - \hat{c}\hat{c}^{\mathrm{T}}, \qquad V_b = \Pi\, V^{\mathrm{lat}}(0)\, \Pi, \tag{28}$$

so that V_b is finite (neutral functions interact as R^{-3}). The two maps between a fitted grid density and the neutral basis are

$$F^{\mathrm{dens}} = M^{-1}\, \Pi\, V_b^{1/2}, \qquad F^{\mathrm{ext}} = M^{-1}\, V^{\mathrm{lat}}(0)\, \Pi\, V_b^{-1/2}, \tag{29}$$

the first for the density side of ε, the second for the potential side of W.

8.3 The response in the charge channel

The constant function couples to the electrons through the transition dipoles. Under periodic boundary conditions the position operator is ill defined, so they are taken in the velocity gauge,

$$\mu^{\alpha}_{ai} = \frac{\langle a|[H, r^{\alpha}]|i\rangle}{\varepsilon_a - \varepsilon_i}, \qquad [H, r^{\alpha}] = -\partial_{\alpha} + [V_{\mathrm{nl}}, r^{\alpha}], \tag{30}$$

and checked against the sum rule

$$\frac{2}{n_{\mathrm{spin}}} \sum_{\sigma} \sum_{ia} 2\,(\varepsilon_a - \varepsilon_i)\, |\mu^{\alpha}_{ai}|^2 = N_{\mathrm{el}}, \tag{31}$$

which a finite basis can only undershoot (h-BN: 0.64/0.64/0.41). The head and the wings of the polarizability in imaginary time are

$$Y_{\alpha\beta}(\mathrm{i}\tau) = \frac{2}{n_{\mathrm{spin}} A} \sum_{\sigma} \sum_{ia} \mu^{\alpha}_{ai} \mu^{\beta}_{ai}\, \mathrm{e}^{-(\varepsilon_a - \varepsilon_i)\tau}, \tag{32}$$

$$X_{\ell\alpha}(\mathrm{i}\tau) = \frac{2}{n_{\mathrm{spin}} \sqrt{A}} \sum_{\sigma} \sum_{ia} \psi_i(\mathbf{r}_\ell) \psi_a(\mathbf{r}_\ell)\, \mu^{\alpha}_{ai}\, \mathrm{e}^{-(\varepsilon_a - \varepsilon_i)\tau}, \tag{33}$$

and are cosine-transformed to imaginary frequency like χ. $Y(\mathrm{i}\omega \to 0)$ is the polarizability per area of the layer (h-BN: 1.35 bohr).

8.4 Inverting ε in the extended basis

With the constant as index 0, the dielectric matrix at small $\mathbf{q}$ is a 2×2 block matrix,

$$\varepsilon(\mathbf{q}, \mathrm{i}\omega) = \begin{pmatrix} h & w^{\dagger} \\ w & B \end{pmatrix}, \qquad h = 1 + 2\pi q\, \hat{\mathbf{q}}^{\mathrm{T}} Y\, \hat{\mathbf{q}}, \qquad w_P = -\mathrm{i}\sqrt{2\pi q}\, (E\hat{\mathbf{q}})_P, \tag{34}$$

$$B(\mathrm{i}\omega) = 1 + F^{\mathrm{dens\,T}} \chi(\mathrm{i}\omega) F^{\mathrm{dens}}, \qquad E(\mathrm{i}\omega) = F^{\mathrm{dens\,T}}\, Z^{\mathrm{T}} X(\mathrm{i}\omega). \tag{35}$$

B is the ordinary Gaussian dielectric matrix at Γ in the neutral basis; h and w are the new head and wings. Inverting the block matrix and keeping the head and the wings of $W^c = V^{1/2}(\varepsilon^{-1} - 1)V^{1/2}$ gives closed forms. All of the $\mathbf{q}$ dependence collapses into one scalar,

$$a(\hat{\mathbf{q}}, \mathrm{i}\omega) = \hat{\mathbf{q}}^{\mathrm{T}} S \hat{\mathbf{q}}, \qquad S = Y - E^{\mathrm{T}} B^{-1} E, \qquad T_{P\alpha} = \left[V^{\mathrm{lat}}(0)\, \Pi\, V_b^{-1/2}\, B^{-1} E \right]_{P\alpha}, \tag{36}$$

(S is the 3×3 head with the body screening folded in, T the wing vector in the auxiliary basis – $MF^{\mathrm{ext}} B^{-1} E$, i.e. in the same representation as W^c of Eq. (23) – both obtained by linear solves with B), and then

$$W^c_{\mathrm{h}}(\mathbf{q}, \mathrm{i}\omega) = -\frac{4\pi^2 a}{1 + 2\pi q a}, \tag{37}$$

$$W^c_{\mathrm{w},P}(\mathbf{q}, \mathrm{i}\omega) = \frac{\mathrm{i}\, 2\pi \sum_{\alpha} \hat{q}_{\alpha} T_{P\alpha}}{1 + 2\pi q a}. \tag{38}$$

Both are finite at $q \to 0$. The head is a number, the wing is a vector over the auxiliary functions and linear in $\hat{\mathbf{q}}$.

8.5 Integral over the Brillouin zone and the correction to W

The head and wings are now integrated over the whole small Brillouin zone of the supercell, with the phase $\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R}}$ of Eq. (27) that turns $W(\mathbf{q})$ into a function of the separation $\mathbf{R} = \mathrm{MIC}(\mathbf{R}_P - \mathbf{R}_Q)$ of the atoms carrying two auxiliary functions. Because a is even and the wing odd in $\hat{\mathbf{q}}$, only cos survives in the head and only sin in the wing, and both results are real:

$$K_{\mathrm{h}}(\mathbf{R}, \mathrm{i}\omega) = \frac{A}{(2\pi)^2} \int_{\mathrm{BZ}} \mathrm{d}^2 q \left[-\frac{4\pi^2 a}{1 + 2\pi q a} \right] \cos(\mathbf{q}\cdot\mathbf{R}), \tag{39}$$

$$K_{\mathrm{w},\alpha}(\mathbf{R}, \mathrm{i}\omega) = \frac{A}{(2\pi)^2} \int_{\mathrm{BZ}} \mathrm{d}^2 q \left[\frac{2\pi\, \hat{q}_\alpha}{1 + 2\pi q a} \right] \sin(\mathbf{q}\cdot\mathbf{R}). \tag{40}$$

The prefactor $A/(2\pi)^2$ makes this the Brillouin-zone *average*, the continuum limit of $\frac{1}{N_k}\sum_k$. The integrals are done in polar coordinates (24 Gauss–Legendre directions per zone edge, 32 radial points), tabulated on a 128×128 mesh of $\mathbf{R}$ over the minimum-image cell and interpolated bilinearly at the atom-pair separations. The correction to the screened interaction in the auxiliary basis is

$$\Delta W_{PQ}(\mathrm{i}\omega) = \frac{c_P c_Q}{A} K_{\mathrm{h}}(\mathbf{R}_{PQ}, \mathrm{i}\omega) + \frac{1}{\sqrt{A}} \sum_\alpha \left[T_{P\alpha} c_Q - c_P T_{Q\alpha} \right] K_{\mathrm{w},\alpha}(\mathbf{R}_{PQ}, \mathrm{i}\omega), \tag{41}$$

symmetric in $P \leftrightarrow Q$ (K_{h} is even and K_{w} odd in $\mathbf{R}$, the wing bracket antisymmetric). It is transformed to imaginary time with the cosine weights of the minimax grid and added to $W^{\mathrm{c}}_{PQ}(\mathrm{i}\tau)$ *before* Eq. (24), so that it reaches the grid pairs through M^{-1} and Z exactly like the regular W. Its head term is a uniform potential $K_{\mathrm{h}}(0)/A$ in real space (-0.29 eV for h-BN). The cost is negligible: one table lookup per auxiliary pair and time point.

8.6 Three-dimensional cells

The same construction applies to a bulk cell (`PERIODIC XYZ`) with three substitutions. The measure is the cell volume, $A \to \Omega$ in Eqs. (27), (32), (33) and (41), and the charge mode is $\phi_0 = \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}}/\sqrt{\Omega}$ with $v_{00} = 4\pi/q^2$.

The Coulomb matrix. The finite box sum of Eq. (13) cannot be used in three dimensions, not even after the subtraction of Eq. (21): between two *neutral* (dipolar) functions the sum of $1/r$ over a 3D box is conditionally convergent, with a surface term $\frac{1}{2} d_\alpha d_\beta \int_{\mathrm{box}} \partial_\alpha \partial_\beta r^{-1}$ that depends on the shape of the box and does not vanish as the box grows (the Lorentz depolarization of a finite sample). The matrix that the head-and-wing model assumes is the tin-foil limit, the periodic Coulomb matrix with the $\mathbf{G} = 0$ Fourier coefficient of $1/r$ omitted, which is obtained as an Ewald sum,

$$V_{PQ}(0) = \sum_{\mathbf{L}\in\mathrm{box}} {}' \left(P\,0 \,\middle|\, \frac{\mathrm{erfc}(\omega r)}{r} \,\middle|\, Q\,\mathbf{L} \right) + \frac{4\pi}{\Omega} \sum_{\mathbf{G}\neq 0} \frac{\mathrm{e}^{-G^2/4\omega^2}}{G^2} \rho_P(\mathbf{G}) \rho_Q(\mathbf{G})^* - \frac{\pi}{\Omega\,\omega^2} c_P c_Q, \qquad \rho_P(\mathbf{G}) = \int \mathrm{d}\mathbf{r}\, \varphi_P(\mathbf{r})\, \mathrm{e}^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}} \tag{42}$$

where $1/r = \mathrm{erfc}(\omega r)/r + \mathrm{erf}(\omega r)/r$, the erf part has the Fourier transform $(4\pi/G^2)\,\mathrm{e}^{-G^2/4\omega^2} = 4\pi/G^2 - \pi/\omega^2 + O(G^2)$, and dropping its $4\pi/G^2$ singularity leaves $-\pi/\omega^2$ at $\mathbf{G} = 0$. The result is independent of ω, which checks both halves against each other; the code sets $\omega r_c = 8$ with r_c the coverage radius of the image box, so that the erfc tail is negligible for every pair the box contains. $\rho_P(\mathbf{G})$ is analytic for Gaussians (one-dimensional moments $m_l(p) = \int \mathrm{d}x\, x^l \mathrm{e}^{-\alpha x^2} \mathrm{e}^{\mathrm{i}px}$ by the recursion $m_l = (\mathrm{i}p\, m_{l-1} + (l-1) m_{l-2})/2\alpha$). This $V(0)$ is finite and positive on the charge direction, so no projector is needed: $\Pi = 1$, $V_b = V(0)$, $F^{\mathrm{dens}} = M^{-1} V^{1/2}$, $G^{\mathrm{ext}} = V^{1/2}$ in Eq. (36), and the regular path of Sec. 7 uses the same matrix with nothing subtracted (the same choice as the 3D branch of cp2k#6447).

Head and wings. In the extended basis the head is $h = 1 + 4\pi\,\hat{\mathbf{q}}^{\mathrm{T}} Y \hat{\mathbf{q}}$ and the wing $w_P = -\mathrm{i}\sqrt{4\pi}\,(E\hat{\mathbf{q}})_P$, both independent of q: the q^2 of the dipole response cancels the $1/q^2$ of v_{00} where in two dimensions $2\pi q\,a$ was left. The block inverse gives, with the same $a = \hat{\mathbf{q}}^{\mathrm{T}} S \hat{\mathbf{q}}$,

$$W^c_{\mathrm{h}}(\mathbf{q},\mathrm{i}\omega) = \frac{4\pi}{q^2}\Big(\frac{1}{1+4\pi a} - 1\Big) = -\frac{4\pi}{q^2}\,\frac{4\pi a}{1+4\pi a}, \qquad W^c_{\mathrm{w},P}(\mathbf{q},\mathrm{i}\omega) = \frac{\mathrm{i}\,(4\pi/q)\sum_\alpha \hat{q}_\alpha T_{P\alpha}}{1+4\pi a}; \tag{43}$$

$1 + 4\pi a(\hat{\mathbf{q}},\mathrm{i}\omega)$ is the macroscopic dielectric constant of the cell along $\hat{\mathbf{q}}$.

Cell integrals. The Brillouin-zone average now carries $\Omega/(2\pi)^3$ and the volume element $q^2 \mathrm{d}q\,\mathrm{d}\Omega_{\hat{\mathbf{q}}}$, which cancels the $1/q^2$ of the head, so that the radial integral is elementary:

$$K_{\mathrm{h}}(\mathbf{R},\mathrm{i}\omega) = -\frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{16\pi^2 a}{1+4\pi a}\int_0^{L(\hat{\mathbf{q}})} \mathrm{d}q\,\cos(q\,\hat{\mathbf{q}}\cdot\mathbf{R}) = -\frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{16\pi^2 a}{1+4\pi a}\,\frac{\sin(Lt)}{t}, \qquad t = \hat{\mathbf{q}}\cdot\mathbf{R}, \tag{44}$$

with $L(\hat{\mathbf{q}})$ the distance from Γ to the boundary of the cell $\{\sum_i s_i \mathbf{b}_i/2,\ |s_i| \le 1\}$ along $\hat{\mathbf{q}}$. The solid-angle integral is a cubature over the six faces of that cell, an 8×8 Gauss–Legendre rule per face with the weight $\mathrm{d}\Omega = |\mathbf{q}\cdot\mathrm{d}\mathbf{S}|/q^3$, checked by $\sum wL^3/3 = (2\pi)^3/\Omega$.

Where the bulk correction is applied: in the orbital basis, not through the RI fit. In two dimensions the head is a ~ 30 meV effect and Eq. (41) can be applied in the auxiliary basis. In three dimensions the head potential $K_{\mathrm{h}}(0,\mathrm{i}\omega)/\Omega$ is of order eV (-3.5 eV for a $3\times3\times3$ diamond cell with $\varepsilon_\infty = 3$) and scales as $1/L$, and the same application fails: $c_P c_Q K_{\mathrm{h}}(\mathbf{R}_{PQ})$ is the projection of a smooth, long-wavelength kernel onto atom-centred Gaussians, its fit through $M^{-1}\cdot M^{-1}$ demands large cancelling coefficients, and the neglected form factors break the cancellation (54-atom diamond: head alone gives a gap of -96 eV and splits a degenerate triplet by 0.4 eV, which a uniform potential cannot do). The correction is therefore evaluated where every term is a linear functional of a pair density. With the extended-basis head $W_{00}(\mathbf{q}) = (4\pi/q^2)(1/\tilde{\varepsilon} - 1)$ and wings $W_{P0}(\mathbf{q}) = \mathrm{i}(4\pi/q)(\hat{\mathbf{q}}\cdot T_P)/\tilde{\varepsilon}$ of Eq. (43), and the small-q expansion of the pair density $\rho_{nm} = \psi_n\psi_m$ against the charge mode,

$$\langle mn|\phi_0\rangle = \frac{1}{\sqrt{\Omega}}\int \mathrm{d}\mathbf{r}\,\rho_{mn}(\mathbf{r})\,\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}} = \frac{\delta_{mn} + \mathrm{i}\,\mathbf{q}\cdot\mathbf{d}_{mn}}{\sqrt{\Omega}} + O(q^2), \qquad \mathbf{d}_{nm} = \langle n|\mathbf{r}|m\rangle = \frac{\langle n|[H,\mathbf{r}]|m\rangle}{\epsilon_n - \epsilon_m}, \tag{45}$$

the Brillouin-zone-cell average of $\langle nm|\Delta W(\mathbf{q})|mn\rangle$ is a real orbital-pair kernel,

$$\hat{W}_{nm}(\mathrm{i}\omega) = w(\mathrm{i}\omega)\,\delta_{nm} + \sum_{\alpha\beta} B_{\alpha\beta}(\mathrm{i}\omega)\,d_{nm,\alpha}d_{nm,\beta} + 2\sqrt{\Omega}\sum_{\alpha\beta} A_{\alpha\beta}(\mathrm{i}\omega)\,V_{nm,\alpha}(\mathrm{i}\omega)\,d_{nm,\beta}, \tag{46}$$

$$w = \frac{K_{\mathrm{h}}(0,\mathrm{i}\omega)}{\Omega}, \quad A_{\alpha\beta} = \frac{1}{(2\pi)^3}\int_{\mathrm{cell}} \mathrm{d}^3q\,\frac{4\pi}{\tilde{\varepsilon}}\hat{q}_\alpha\hat{q}_\beta, \quad B_{\alpha\beta} = \frac{1}{(2\pi)^3}\int_{\mathrm{cell}} \mathrm{d}^3q\,4\pi\Big(\frac{1}{\tilde{\varepsilon}} - 1\Big)\hat{q}_\alpha\hat{q}_\beta, \tag{47}$$

$$V_{nm,\alpha}(\mathrm{i}\omega) = \int \mathrm{d}\mathbf{r}\,\rho_{nm}(\mathbf{r})\,w_\alpha(\mathbf{r};\mathrm{i}\omega) = \sum_\ell \rho_{nm}(\mathbf{r}_\ell)\,\big[ZM^{-1}T(\mathrm{i}\omega)\big]_{\ell\alpha}, \qquad w_\alpha = \sum_P \hat{\varphi}_P\,T_{P\alpha}, \tag{48}$$

where the head is a uniform potential (order $1/L$), the head-dipole term is the $O(q^2)$ remainder of the head, and the wing couples the dipole of the pair density to the potential w_α that the uniform field induces in the body ($A, B \propto 1/\Omega$ and $T \propto 1/\sqrt{\Omega}$, so both are of order $1/\Omega$). The wing carries a single $1/\sqrt{\Omega}$ from Eq. (45) against the $\Omega/(2\pi)^3$ of the average, hence $2\sqrt{\Omega}$; the head terms carry $|\langle mn|\phi_0\rangle|^2 \propto 1/\Omega$, which the average cancels. V_{nm} is a linear functional of ρ_{nm} and is exact on the RI-RS grid, unlike the pair kernel of Eq. (41). Degenerate pairs, where Eq. (45) is not defined, are set to zero; between folded states of different crystal momentum the commutator vanishes identically. With Eqs. (53)–(55) and $\hat{W}$ transformed to imaginary time with the cosine weights of the minimax grid,

$$\Delta\Sigma^c_n(-\mathrm{i}|\tau|) = -\sum_{m\in\mathrm{occ}} \mathrm{e}^{-|\tau|(\varepsilon_F - \epsilon_m)}\hat{W}_{nm}(\mathrm{i}\tau), \qquad \Delta\Sigma^c_n(+\mathrm{i}|\tau|) = +\sum_{m\in\mathrm{vir}} \mathrm{e}^{-|\tau|(\epsilon_m - \varepsilon_F)}\hat{W}_{nm}(\mathrm{i}\tau), \tag{49}$$

added to the stored orbital matrices as $(SC)\,\mathrm{diag}(\Delta\Sigma_n)\,(SC)^{\mathrm{T}}$ so that it folds onto the band and DOS k-points like the regular Σ^c ($C^{\mathrm{T}}SC = 1$). For the head alone this is $\mp w(\mathrm{i}\tau)\,\mathrm{e}^{-|\tau(\epsilon_n-\varepsilon_F)|}$ on the diagonal, the $\pm w/2$ shift of the COHSEX head in the static limit. No grid kernel and no fit of a kernel is involved; the cost per frequency is three $\Phi^{\mathrm{T}}\mathrm{diag}(\tilde{w})\Phi$ products on the grid, six $n_{\mathrm{AO}}^2 n_{\mathrm{MO}}$ transforms and an n_{MO}^2 element loop.

9 Exchange self-energy

The exchange part uses the truncated Coulomb operator v_{tc} with a cutoff radius equal to half the minimum cell dimension (15.2 Å for the h-BN cell), which is finite without any treatment of the charge channel and is unaffected by everything above:

$$P_{\mu\nu} = \sum_{i}^{\mathrm{occ}} C_{\mu i} C_{\nu i}, \qquad P_{\ell\ell'} = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell) P_{\mu\nu} \Phi_\nu(\mathbf{r}_{\ell'}), \tag{50}$$

$$V^{\mathrm{tc}}_{\ell\ell'} = \sum_{PQ} Z_{\ell P} \left[M^{-1} V^{\mathrm{tc}} M^{-1}\right]_{PQ} Z_{\ell' Q}, \tag{51}$$

$$\Sigma^x_{\lambda\sigma} = -\sum_{\ell\ell'} \Phi_\lambda(\mathbf{r}_\ell)\, P_{\ell\ell'}\, V^{\mathrm{tc}}_{\ell\ell'}\, \Phi_\sigma(\mathbf{r}_{\ell'}). \tag{52}$$

10 Correlation self-energy and quasiparticle energies

The correlation self-energy is an element-wise product on the grid of the Green's function with the *full* screened interaction, the regular part of Eq. (25) plus the charge-channel correction of Eq. (41), in the two time branches:

$$\Sigma^c_{\ell\ell'}(-\mathrm{i}|\tau|) = -\,G^{\mathrm{occ}}_{\ell\ell'}(\mathrm{i}|\tau|)\left[W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) + \Delta W_{\ell\ell'}(\mathrm{i}\tau)\right], \tag{53}$$

$$\Sigma^c_{\ell\ell'}(+\mathrm{i}|\tau|) = +\,G^{\mathrm{vir}}_{\ell\ell'}(\mathrm{i}|\tau|)\left[W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) + \Delta W_{\ell\ell'}(\mathrm{i}\tau)\right], \tag{54}$$

$$\Sigma^c_{\lambda\sigma}(\pm\mathrm{i}|\tau|) = \sum_{\ell\ell'} \Phi_\lambda(\mathbf{r}_\ell)\, \Sigma^c_{\ell\ell'}(\pm\mathrm{i}|\tau|)\, \Phi_\sigma(\mathbf{r}_{\ell'}). \tag{55}$$

In the implementation $W^{\mathrm{MIC}}_{\ell\ell'}$ is built panel by panel as $Z\,W^{\mathrm{MIC}}_{PQ}\,Z^{\mathrm{T}}$, with ΔW already inside W^{MIC}_{PQ} (Eq. (41)), and the two Hadamard products with G^{occ} and G^{vir} share that panel. The panels are restricted to the pairs within the geometric cutoff of the Σ^c panels, which is exact because the Hadamard product with the short-ranged G vanishes outside it.

The self-energy is projected onto the Kohn–Sham states, $\Sigma^c_{nn}(\pm\mathrm{i}|\tau|) = \sum_{\lambda\sigma} C_{\lambda n}\Sigma^c_{\lambda\sigma}(\pm\mathrm{i}|\tau|)C_{\sigma n}$, split into its even and odd parts in τ and transformed to imaginary frequency with the cosine and sine weights of Eq. (3),

$$\Sigma^c_{nn}(\mathrm{i}\omega_k) = \sum_j \gamma^{\cos}_{kj}\,\frac{\Sigma^c_{nn}(\mathrm{i}\tau_j) + \Sigma^c_{nn}(-\mathrm{i}\tau_j)}{2} + \sum_j \gamma^{\sin}_{kj}\,\frac{\Sigma^c_{nn}(\mathrm{i}\tau_j) - \Sigma^c_{nn}(-\mathrm{i}\tau_j)}{2}. \tag{56}$$

$\Sigma^c_{nn}(\mathrm{i}\omega)$ is continued to the real axis by a Padé approximant fitted to the imaginary-frequency points, and the quasiparticle energies follow from

$$\varepsilon^{GW}_n = \varepsilon_n + \mathrm{Re}\,\Sigma^c_{nn}(\varepsilon^{GW}_n) + \Sigma^x_{nn} - v^{xc}_{nn}, \tag{57}$$

solved by a Newton search for the crossing of the quasiparticle equation. For G_0W_0 this is done once with G built from the Kohn–Sham eigenvalues; for evGW_0 the Green's function is rebuilt from the quasiparticle energies and Eqs. (10)–(56) are iterated to self-consistency with W frozen.

11 Validation

Table 1 collects the tests. The water cells are deliberately far from the regime of the method — a 6 Å cell has a Brillouin zone large enough that the body of W disperses across it, and that dispersion is not corrected by a Γ-only evaluation — but they have an exact reference (the shifted even meshes converge to 23.25 eV) and run in a minute. The residual error halves from the 6 Å to the 12 Å cell, as a body-dispersion error must. For the h-BN supercell the 2×2 and 4×4 meshes agree to 13 meV, so the body is flat; there the Γ-only result is 0.1 eV from the converged value, and the residual is a single uniform shift of ∓ 0.052 eV of every occupied/empty Σ^c (standard deviation 3 and 18 meV over all 3136 states) with exact degeneracies — one small constant in the charge-channel W^c. It is not the box edge of Eq. (21): raising the lattice-sum level from $s = 4$ to $s = 5$ (three times larger box, D_{box} 1.52 → 4.57, removed constant $7.4 \times 10^4 \to 2.2 \times 10^5$ Ha) leaves the remainder ($-139 \to -130$ Ha), the gap (6.699 → 6.706 eV) and the per-state offset unchanged. What remains are the approximations of the model itself: the body of W is taken at Γ only, and the constant-function head carries no $\mathrm{e}^{-q|z-z'|}$ form factor for the finite thickness of the layer.

system	cell	Γ-only + corr.	shifted-mesh reference	before
H_2O	6×6 Å	22.789 (s=3), 22.855 (s=4)	23.262 (2×2), 23.248 (4×4)	15.751
$2\,H_2O$	12×6 Å	23.037	23.251 (2×2)	11.063
h-BN	14×14, 392 atoms	**6.699** (s=4), 6.706 (s=5)	6.595 (2×2), 6.582 (4×4)	33.116 / −31.844

Table 1: G_0W_0 gaps in eV. The h-BN Γ-only calculation took 1060 s against 1665 s for the 2×2 production setting on the same resources; of that, the charge-channel fit is 131 s and the ΔW kernel 47 s. "Before the fixes" refers to the same Γ-only input with (i) the hidden 2×2 extrapolation mesh still active, (ii) the neutral projection $\Pi V \Pi$ in the W path instead of Eq. (21), and (iii) sphere-local fitting of $Z_{\ell 0}$; the two h-BN numbers are with the length-gauge and the velocity-gauge dipoles, respectively.

Summary of the evaluation order

1. $\Phi_\mu(\mathbf{r}_\ell)$, Eq. (4); $Z_{\ell P}$ sphere by sphere, Eqs. (5)–(8).
2. $G^{\mathrm{occ}}, G^{\mathrm{vir}}$, $\chi_{\ell\ell'}$, $\chi_{PQ}(\mathrm{i}\tau) \to \chi_{PQ}(\mathrm{i}\omega)$, Eqs. (10)–(12).
3. c_P, Π, V_b, F^{dens}, F^{ext}, Eqs. (14)–(29); dipoles and sum rule, Eqs. (30)–(31); Y, X, B, E, S, T, Eqs. (32)–(36); tables K_{h}, K_{w}, Eqs. (39)–(40); c_P, Eq. (27).
4. $V^{\mathrm{lat}}(0) \to V^{\mathrm{reg}}(0)$, Eqs. (13)–(21) (3D: the Ewald $V(0)$ of Eq. (42), nothing subtracted; the head potential of Eq. (44) goes into Σ^c by Eq. (49)); ε, W^c, $W^c(\mathrm{i}\omega) \to W^c(\mathrm{i}\tau)$, $W^c_{PQ}(\mathrm{i}\tau)$ += $\Delta W_{PQ}(\mathrm{i}\tau)$, then W^{MIC}, Eqs. (22)–(24), (41).
5. Σ^x, Eq. (52).
6. Per time point and panel: $W^{\mathrm{MIC}}_{\ell\ell'}$, Eq. (25); Σ^c in both branches, Eqs. (53)–(55); then Eqs. (56)–(57).